SAMPLE

Worlds Beyond Earth: Callisto

The Cratered Moon

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Introduction

Of all the worlds the solar system offers, the strangest category may be the ocean moons: bodies far from the Sun, frozen on the outside, and yet carrying, under their icy crusts, more liquid water than all the Earth's oceans hold. Jupiter's outermost great moon, Callisto, is one of these, and it is the oddest of the family: a world as large as the planet Mercury, more heavily cratered than any other body its size, ancient and quiet at the surface, and yet, by the gravity measurements, layered inside like an ocean moon, a possible sea of salt water lying deep beneath a shell of rock and ice four times older than the Earth's continents.

The moon's surface is the solar system's oldest landscape. Nothing dramatic has happened to Callisto for four billion years: no volcanism has repaved it, no tectonics have wrinkled it, no atmosphere has weathered it. The record of the solar system's violent youth, on the Earth long erased, on the Moon partly erased, on Callisto survives almost whole, the craters of the late bombardment and earlier still sitting where they fell, a fossil surface, the book's central subject, the closest thing the solar system keeps to a photograph of its own childhood.

The moon's interior is the opposite surprise. The same spacecraft that revealed the ancient surface revealed the modern sea: gravity measurements made during the Jupiter orbiter's repeated Callisto flybys showed the moon's interior is partly decoupled, the outer shell not locked to the deep rock, the signature of a layer of liquid, and the best current reading puts a salty ocean tens of kilometers down, kept liquid not by the Sun's weak warmth at Jupiter's distance but by the moon's own slow radioactivity and, perhaps, the trace of the tidal kneading its slightly stretched orbit still provides. The reader should hold the combination as the book's organizing wonder: the solar system's oldest face over one of its newest discoveries, the fossil and the sea in one body.

The book's chapters follow the story in order: the discovery, in the seventeenth century, by the same Italian observer who found Jupiter's other great moons; the moon's character, size and heft and the strange absence of any magnetic personality of its own; the interior, ocean and rock, and the evidence that found it; the exploration, from the two Voyager snapshots to the Jupiter orbiter's three dozen passes; the surface, the crater record and what it teaches about the solar system's youth; the moon's place among the Galileans, Jupiter's family of four; and the human prospects, the radiation shelter that makes Callisto the proposed base camp of the Jupiter system, and the far-future dream of the ocean's exploration, the whole story of a moon that was, for three centuries, the sky's most boring large body and is now among its most interesting.

The introduction's reader should hold the book's honest boundaries, because the moon's story sets them. The ocean is measured but unmapped, the interior is modeled but unprobed, the future chapters describe plans rather than programs, and the reader will find, throughout, the field's own phrasing: inferred, modeled, candidate, proposed. The book's certainty lies in the old facts, the discovery, the size, the craters, the orbit, and the reader can trust those completely, the moon being the rare subject whose public facts are its stable ones, and the frontier facts being labeled as frontier, the reader being given, as the campaign's habit promises, the science's own confidence levels, stated plainly, chapter by chapter, so that the wonder at the end is the earned kind.

The reader should hold the moon's one-line biography, because the chapters need a peg. Callisto: diameter 4,800 kilometers, orbit 1.9 million kilometers, period seventeen days, density 1.83, surface age four billion years, ocean probable, magnetic field absent, radiation safe, the moon being the solar system's third-largest satellite and its least dramatic, the biography's facts being the book's chapters in miniature, the reader who memorizes the one line holding the skeleton every chapter fills, the numbers being the moon's address, the story being what happened there, the seventh sentence of the family's song and the quietest.

The biography's reader should note the moon's discoverable-with-binoculars status, because the amateur connection is part of the story. Callisto is visible from a dark backyard in good binoculars, the four Galileans being the sky's standard binocular prize, and the fourth moon, at magnitude five, rewards the steady hand, the moon appearing as a star that shifts night to night along Jupiter's side, the patient watcher learning its seventeen-day beat in a week of evenings. The reader should feel the lineage's point: the moon that the pamphlet made famous is, tonight, findable by anyone, the discovery being repeatable with a hundred-dollar instrument, the moons having never stopped being the public's own astronomy, the fourth moon being a sight the reader can check personally, the book's first verification being an evening's project, the sky having kept the appointment for four hundred years.

The amateur reader should learn the fourth moon's observing rhythm, because the moon rewards patience with pattern. Callisto's seventeen-day circuit swings the moon from Jupiter's east to its west and back, the moon's greatest elongations, the best viewing, coming at the swing's ends, and the moon, alone of the four, can pass

above or below Jupiter rather than in front, the geometry the system's tilt permits, the fourth moon's seasons having the long beats, the reader who tracks the moon for a month seeing the whole cycle, the pattern the pamphlet's author saw in the winter of 1610, the four hundred years between his nights and the reader's being a continuity the moon's own patience earns, the observation being the same observation, the instrument being the only change.

The rhythm's reader should try the moon's identification test, because the four dots reward the careful eye. The Galileans, strung along Jupiter's equator, are distinguishable by brightness and spacing, the inner moons moving fast and close, the outer slow and far, the fourth moon being the faintest and the farthest flung, its dot the hardest-won, the reader's test being the week's log: plot the dots nightly, and the pattern sorts itself, the two inner lights racing, the third stately, the fourth crawling, the moon's own identity emerging from the arithmetic of patience, the reader learning in a fortnight what the pamphlet's winter taught the world, the sky's oldest visible clockwork still teaching the same lesson to anyone who keeps the notebook.

The notebook's reader should close the amateur chapter with the eclipse hunt, because the fourth moon offers one. Callisto's shadow, in the transit seasons, crosses Jupiter's face, the outer moon's transit being the rarest of the four, the dot of shadow taking hours to cross, the patient backyard watcher with a modest scope catching the event the almanacs list, the season's dates being published, the reader's project being the winter's watch. The reader should feel the hunt's lineage: the shadow the Danish astronomer timed, the delay that measured light's speed, is the same shadow the backyard sees, the eclipse clock's hands still turning, the amateur's timing being the descendant of the founding measurement, the moon's public service, four centuries long, being simply this: the moon keeps showing up, and the sky's arithmetic keeps working, the quiet satellite being the amateur's steady friend, the fourth dot's patience rewarding the reader's own.

Chapter 01: The Discovery: Galileo's Fourth Moon

The moon's discovery belongs to the founding week of observational astronomy, and this chapter is the discovery: the Italian observer's 1610 campaign, the fourth moon's place in the sequence, and the name, a mythological scandal, that the moon ended up wearing.

The discovery's setting gets the reader's picture. In January 1610, the Pisan mathematician, then a professor at Padua, turned his improved telescope toward Jupiter and found, night after night, small stars lined up near the planet, stars that moved, and moved in step with the planet, and switched sides, the arrangement unmistakable: Jupiter carried attendants, three of them at first, then four, the fourth and outermost faint and slow. The reader should feel the finding's weight in its time: the doctrine taught that everything circled the Earth, and here, visible in a tube anyone could build, were four bodies circling something else, the first clear crack in the Earth-centered cosmos, and the fourth moon, the outermost and faintest of the four, was part of the opening night's evidence, the discovery that helped end an astronomy two thousand years old.

The discovery's spread takes a paragraph of its own, because the fourth moon entered the world's argument quickly. The little book the discoverer published that spring, the pamphlet with the two-word title, carried the nightly drawings of Jupiter's attendants, the dots marching, and the pamphlet reached the thinkers of the day within months, the reaction splitting between the converted and the hostile, the celestial establishment's refusal running on every ground from scripture to eyesight, the one charge the defenders answering easily, the telescopes being makeable by anyone. The reader should feel the discovery's place in the larger fall: the four moons were the second blow, the phases of Venus being the first, and the fourth moon, faintest and slowest, being the quietest part of the evidence, the observer's tables giving the doubters nothing to work with, the measurements being what they were, the moons being where the tables said, every night, for anyone who looked.

The fourth moon's own discovery receives its precision, because the record is recoverable. The observer's notes of the first week show three objects, the third and fourth confused at first, their positions uncertain by a night or two, and the fourth, the outermost, established as a distinct body within the first days of the campaign, its period, about seventeen days, the longest of the four, its brightness the least. The reader should note the moon's immediate handicap: faintest and outermost, the fourth moon was the hardest of the four for every early telescope, its observations the sparsest in the discovery literature, the first member of the family to be mistaken for a star and the last to be measured well, a distinction of dimness the moon would carry for three centuries, its story, more than the others', being the story of what patience and good instruments eventually find.

The name is the discovery's scandal, and the reader should enjoy it knowingly. The discoverer himself, cautious, named the bodies the Medicean stars, for his patrons, and it was another astronomer, the German Simon Marius,

claiming an independent discovery a few weeks later, who gave the moons the mythological names they now wear, and the fourth, outermost moon received the name of Callisto, a nymph of the Greek myths, an attendant of the goddess Artemis, whose story in the ancient telling is a grim one: a victim of the king of the gods' pursuit, transformed, exiled, and finally set in the sky as the bear. The reader should hold the naming's odd fitness by accident: the moon called after a figure transformed from companion to exile bears, of the four, the most ancient face and the farthest camp, the name's darkness and distance suiting the moon that the next chapters show to be the family's quiet survivor.

The naming's aftermath merits a closing note, because the modern reader inherits the dispute. The discoverer, for reasons of prudence and patrons, kept the Medicean names in his own writings, the mythological names living in the other discoverer's book and in the colloquial usage, the two sets competing for a century, and the myths winning, the myths being the people's choice, the reader should note, the bodies entering the permanent record with the names that scandalized the first readers. The reader may enjoy the footnote that the German claimant, whose priority was doubted then and is doubted now, ended up supplying the solar system's most durable vocabulary, the moons of Jupiter wearing his myths, and the fourth moon, the exile's name, being the most mythological of all, the sky's most literal exile, the farthest and the quietest, the name being the one item of the moon's biography that is pure story.

The naming card shuts the discovery chapter's ledger. The fourth moon received the name that would eventually christen the whole class, the planet's mortal companion, and the file notes the naming's quiet weight: a figure the mythology records as punished, attached forever to the moon that became the standard candle of outer-system time. The reader should take the entry at its honest size, the name, mythologically exact, the fate, dropped from the moon's own card, and the naming, the discovery file's own footnote, kept.

The slow-motion card prices the discovery's own temper. The fourth moon was the last of the lot to enter the notebook, found in a week of tilted fortunes, and the file's own note is that the discovery was nearly missed twice over: the moon is faint, the night was imperfect, and the observation's worth rested on the watcher's discipline rather than the sky's generosity. The reader should take the card as written, the catch, late in the week, the worth, earned by patience, and the slow motion, the discovery file's own tempo, kept.